We now work with the **full i.i.d. sample likelihood**:

$$\ell^{X_t}(\mu, \sigma) = \sum_{t=1}^{T} \log f(x_t; \mu, \sigma)$$

under the **Gumbel (location–scale)** distribution.

✅ 1. Log-likelihood under Gumbel

Recall:

$$f(x; \mu, \sigma) = \frac{1}{\sigma} \exp\left(-\frac{x-\mu}{\sigma}\right) \exp\left(-e^{-(x-\mu)/\sigma}\right)$$

Define:

$$z_t = \frac{x_t - \mu}{\sigma}$$

Then:

Log-density:

$$\log f(x_t) = -\log \sigma - z_t - e^{-z_t}$$

✅ Final log-likelihood

$$\boxed{\ell(\mu, \sigma) = -T \log \sigma - \sum_{t=1}^{T} z_t - \sum_{t=1}^{T} e^{-z_t}}$$

or equivalently:

$$\boxed{\ell(\mu, \sigma) = -T \log \sigma - \frac{1}{\sigma} \sum_{t=1}^{T} (x_t - \mu) - \sum_{t=1}^{T} \exp\left(-\frac{x_t - \mu}{\sigma}\right)}$$

✅ 2. First derivatives (Gradient)

✅ Derivative w.r.t. μ

Using:

$$\frac{\partial z_t}{\partial \mu} = -\frac{1}{\sigma}$$

$$\boxed{\frac{\partial \ell}{\partial \mu} = \frac{T}{\sigma} - \frac{1}{\sigma}\sum_{t=1}^{T} e^{-z_t}}$$

✅ Derivative w.r.t. σ

Using:

$$\frac{\partial z_t}{\partial \sigma} = -\frac{x_t - \mu}{\sigma^2}$$

$$\boxed{\begin{aligned}\frac{\partial \ell}{\partial \sigma} = -\frac{T}{\sigma} + \frac{1}{\sigma^2}\sum_{t=1}^{T}(x_t - \mu) \\ - \frac{1}{\sigma^2}\sum_{t=1}^{T}(x_t - \mu)e^{-z_t}\end{aligned}}$$

✅ 3. Second derivatives (Hessian)

✅ Second derivative w.r.t. μ

$$\boxed{\frac{\partial^2 \ell}{\partial \mu^2} = -\frac{T}{\sigma^2} - \frac{1}{\sigma^2}\sum_{t=1}^{T} e^{-z_t}}$$

✅ Second derivative w.r.t. σ

$$\boxed{\begin{aligned}\frac{\partial^2 \ell}{\partial \sigma^2} = \frac{T}{\sigma^2} - \frac{2}{\sigma^3}\sum_{t=1}^{T}(x_t-\mu) \\ + \frac{2}{\sigma^3}\sum_{t=1}^{T}(x_t-\mu)e^{-z_t} \\ + \frac{1}{\sigma^4}\sum_{t=1}^{T}(x_t-\mu)^2 e^{-z_t}\end{aligned}}$$

✅ Cross derivative

$$\boxed{\begin{aligned}\frac{\partial^2 \ell}{\partial \mu \partial \sigma} = -\frac{T}{\sigma^2} + \frac{1}{\sigma^2}\sum_{t=1}^{T}e^{-z_t} \\ - \frac{1}{\sigma^3}\sum_{t=1}^{T}(x_t-\mu)e^{-z_t}\end{aligned}}$$

✅ 4. Summary (compact form)

Let:

$$e_t = e^{-z_t}, \quad y_t = x_t - \mu$$

Then:

Gradient

$$\begin{aligned}\ell_\mu &= \frac{T}{\sigma} - \frac{1}{\sigma}\sum e_t \\ \ell_\sigma &= -\frac{T}{\sigma} + \frac{1}{\sigma^2}\sum y_t - \frac{1}{\sigma^2}\sum y_t e_t\end{aligned}$$

Hessian

Hessian

$$
\begin{aligned}
\ell_{\mu\mu} &= -\frac{T}{\sigma^2} - \frac{1}{\sigma^2}\sum e_t \\
\ell_{\sigma\sigma} &= \frac{T}{\sigma^2} - \frac{2}{\sigma^3}\sum y_t + \frac{2}{\sigma^3}\sum y_t e_t + \frac{1}{\sigma^4}\sum y_t^2 e_t \\
\ell_{\mu\sigma} &= -\frac{T}{\sigma^2} + \frac{1}{\sigma^2}\sum e_t - \frac{1}{\sigma^3}\sum y_t e_t
\end{aligned}
$$